



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2023-24(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: V

Programme: B. Tech Information Technology

Course: Data Warehousing and Mining

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33505	Course Outcome
CO1	To classify the fundamentals, need, structure, and cost of data warehousing for effective data management.
CO2	To classify data warehousing architecture, modeling techniques, ETL processes, and OLAP operations for efficient data analysis.
CO3	To explain data preprocessing techniques and transformation for improving data quality.
CO4	To provide fundamental knowledge of data mining concepts, techniques, and applications.
CO5	To introduce classification techniques and equip students with the ability to compare and apply different classifiers effectively.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define Data Warehouse.	CO1	2	2
b.	What is star schema?	CO2	2	2
c.	Define data preprocessing.	CO3	3	2
d.	What is data mining?	CO4	2	2
e.	Define Big Data.	CO5	2	2

Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain need and characteristics of data warehouse.	CO1	2	5
b.	Compare OLTP and OLAP systems.	CO1	2	5
c.	Explain structure and components of data warehouse.	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain ETL process in detail.	CO2	2	5
b.	Compare star, snowflake, and fact constellation schemas.	CO2	3	5
c.	Explain OLAP operations with examples.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain data preprocessing steps.	CO3	4	5
b.	Discuss data cleaning and integration techniques.	CO3	3	5
c.	Explain data transformation strategies.	CO3	3	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain data mining tasks and applications.	CO4	2	5
b.	Discuss issues in data mining.	CO4	4	5
c.	Explain relationship between data warehousing and data mining.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain BI architecture and applications.	CO5	2	5
b.	Discuss Big Data challenges and Hadoop ecosystem.	CO5	3	5
c.	Explain development of BI system.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create